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Multimode cable vibration control using a viscous-shear damper: Case studies on the Sutong Bridge

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Summary

This study investigates multimode damping effects of a long cable attached with a viscous-shear damper (VSD). A typical VSD comprises a casing box containing viscous medium and shearing plates with parts submerged in the medium. It dissipates vibration energy through shear deformation of the viscous medium. The VSD is low-cost and invulnerable to leakage or increasing joint play and hence requires low maintenance effort. For studying its damping effects on long cables, three VSDs were designed respectively for three long cables of the Sutong Bridge, a cable-stayed bridge with a main span of 1,088 m, and then tested in laboratory under forced sinusoidal displacements with various frequencies and amplitudes. Their dynamic behaviors were found to be accurately modeled using a viscous damper with intrinsic stiffness, while the viscous and stiffness coefficients depend on frequency and amplitude. The VSDs were subsequently attached to the cables respectively, and modal damping ratios of in-plane modes of each cable without dampers and with the VSD were measured by free-decay testing. The measurements obtained for most in-plane modes with frequency less than 3 Hz show that the VSD provides satisfactory multimode damping effects with efficiency factors between 30% and 50% as compared to an ideal viscous damper and the damping decreases slightly for higher modes. The measurements are also found consistent with the results of complex modal analysis of a shallow cable with a Kelvin–Voigt damper considering frequency- and amplitude-dependent damper properties. Particularly, analytical prediction using the damper properties tested under small deformation amplitude provides the lower bound of the damping effect.

KEYWORDS

cable vibration control, experimental study, field testing, multimode damping effect, viscoelastic damper, viscous-shear damper

1 | INTRODUCTION

Nowadays, problematic vibrations of stay cables in cable-stayed bridges are mainly suppressed by aerodynamic surface treatments of cable pipes, installing a high-damping rubber damper internally in cable anchorage guide pipe, adding external dampers between cable and bridge deck, or using cross-ties.^{1,2} For long cables of long-span bridges, two or

more of these countermeasures are commonly applied in practice, for example, the cable vibration control system on the Fred Harten Bridge,¹ the Stonecutters Bridge,³ and the Sutong Bridge.⁴

Among the countermeasures against cable vibrations, installation of external/internal dampers is one of the most effective and well-studied ones.⁵ The damping effect of a taut cable with an ideal viscous damper was first obtained by Pacheco et al⁶ for an arbitrary mode and an arbitrary damper location near cable anchorage, using a mode-superposition-based model with complex eigenvalue analysis. For this nonclassically damped system, the analytical method, that is, the complex modal analysis, has been later established.⁷⁻⁹ Effects of the sag of long cables and the cable bending stiffness were subsequently studied.¹⁰⁻¹³ The performance of real dampers, whose properties normally depart from those of an ideal viscous damper, on cables has attracted more attention after an experimental measurement campaign carried out by Sun et al^{14,15} in 2002, where various types of dampers including the oil damper, the hydraulic damper, the friction damper, the shear-type viscous damper, and the Magneto-Rheological (MR) damper were respectively attached to a real-scale cable for testing. The results thereby show that damper properties including intrinsic stiffness and support stiffness have significant influences on the achievable damping. Complex modal analysis to understand the influences of such factors was further carried out.¹⁶⁻¹⁹ Notable is that the damper stiffness had already been considered in 1990s by Yoneda^{20,21} when studying viscous-shear dampers (VSDs) for cables, using complex eigenvalue analysis and simplified methods though. A comprehensive analysis of a sagged cable with bending stiffness attached with a viscous damper considering support flexibility or a high damping rubber was performed by Fujino and Hoang,²² and the damper stiffness was further considered by Javanbakht et al.²³ Apparently, the cable-damper system has been extensively studied using numerical and analytical methods in the last decades. Recent studies tend to focus on enhancing the damping effect of near-anchorage dampers to compensate for the inadequate supplemental damping for long cables due to the limited damper installation distance from the cable anchorage and the sag effect, for example, using inerter or negative stiffness mechanism and so on²⁴⁻³⁰ and characterizing complex cable networks with cross-ties.³¹⁻³⁶ Meanwhile, numerical methods have been applied to precisely appreciate the influences of damper nonlinearity on the damping effects.^{37,38}

Beside the theoretical analysis, experimental studies have been carried out on cable-damper systems. Considering the large length of stay cables, most experiments were performed using scaled models of the cable and the damper, for example, for qualitative validation of theoretical findings^{6,18} and for developing novel devices or control algorithms.^{25,39} There have been several experiments carried out on real-scale cable-damper systems, to quantitatively evaluate the damping effects of particular dampers, for example, using a prototype cable of 228 m of the Sutong Bridge, China, with corresponding results reported by Sun et al^{14,15} and Zhou et al,^{40,41} and using a cable of 168.25 m.⁴² Field measurements of damper performances have also been reported in the literature. Mostly, they were associated with damper design for specific cable-stayed bridges. For examples, in 1990s, the VSDs were studied and tested^{20,21,43,44} on the Sakitamao Bridge in Toda, Japan, and the Koshikidaimyojin Bridge in Satsumasendai, Japan. The performance of a MR damper with novel control algorithms was tested on the longest cable (163.7 m) of the Eiland Bridge near Kampen, the Netherlands, and on a cable of 317 m of the Russky Bridge near Vladivostok, Russia, as reported by Weber et al.⁴ The damping effects of hydraulic dampers on cables of the Stonecutter Bridge were tested before its opening.^{3,45} Comprehensive field vibration tests were performed on a 115-m cable of the Dongting Lake Bridge in China, to study the damping effects of MR dampers with different voltage strength,⁴⁶ where the first 20 vibration modes were tested and the amplitude-dependent damping effects were analyzed. Cable damping measurements have also been reported for the Leonard P. Zakim Bunker Hill Bridge¹ before installation of dampers and cross-ties, after damper installation, and after cross-tie installation, respectively, where the first three modes were tested by manual excitations. The performance of viscous dampers on the Fred Harten Bridge was evaluated from monitored cable responses before and after the damper installation based on the statistics of cable responses.⁴⁷

Note that long span bridges are generally designed to have a service life of 100 years or even longer and the cables are supposed to serve for more than 50 years. However, cable dampers normally have a shorter service life, roughly of 10 years, since they are subjected to dynamic loading due to cable vibrations and quasi-static loading owing to traffic and temperature-induced deck deflection.⁴⁸⁻⁵⁰ Therefore, it is of practical significance to develop low-cost dampers with low-maintenance requirements. Considered here is the VSD by which the vibration energy is dissipated through shear deformation of a viscous medium. The VSD has a simple structure and is low-cost.²⁰ Besides, it is not subjected to the problems encountered by fluid viscous dampers or oil dampers including leakage and increasing joint play in long-term use. The VSD has been studied in Japan in 1990s^{20,21,43,44} and has applications in both Japan and China, mainly for cables of small to medium length. In this study, VSDs are investigated in detail by analytical analysis, laboratory experiments, and field tests on long cables of the Sutong Bridge, which is a cable-stayed bridge in China, with a main span of

1,088 m and has been opened for 10 years. Previous full-scale measurements, for example, on the Fred Hartman Bridge in Houston, Texas, have shown that most of the rain–wind-induced responses have frequencies between 1 and 3 Hz.⁵¹ Therefore, multimode vibration control effects are important for practical design.⁵² Particularly, for long cables whose fundamental frequency is quite low, the dominant mode in rain–wind-induced cable vibrations could range from 1 to over 10. This study hence focuses on the damping effects for all the cable modes with frequency lower than 3 Hz.

This paper is structured as follows. Section 2 briefly describes the Sutong Bridge and the cables for study. Section 3 introduces the VSDs and the laboratory testing. The procedure and the results of the field testing of these dampers on the cables are presented in Section 4 along with discussion. The findings are briefly summarized in Section 5.

2 | THE SUTONG BRIDGE AND THE STAY CABLES FOR STUDY

The Sutong Bridge is located in Jiangsu Province, China, connecting Nantong and Suzhou, over the Yangtze River. It was completed and opened to traffic in May 2008. The main navigation channel bridge is a cable-stayed bridge with a total length of 2,088 m and a main span of 1,088 m, as illustrated in Figure 1, which will be referred to ambiguously as the Sutong Bridge hereafter. The Sutong Bridge has a total of 272 stay cables with length between 153 and 577 m. Aero-dynamic surface treatments, internal rubber dampers, and external dampers have been implemented for cable vibration control.⁴

Three long cables of different length were selected for design and field testing of the VSDs. They are indicated by red thick lines in Figure 1. The cable parameters and basic dynamic properties are listed in Table 1. In the table, the three cables are numbered according to their length, cable length is denoted by l , cable mass per unit length is denoted by m , D is cable diameter, and cable tension is denoted by H . The sag-extensibility parameter λ^2 is computed as⁵³

$$\lambda^2 = \left(\frac{mgl \cos \theta}{H} \right)^2 \frac{EA l}{HL_e}, \quad (1)$$

where g = gravitational acceleration, θ = cable inclination angle, EA = cable axial stiffness, and E = the modulus of elasticity of cables ($E = 200$ GPa in this study). The stretched length of the cable in static state is denoted by L_e .

Taking into account the sag effect, the cable frequency is computed using the linear theory for shallow cables,^{53,54} as will be further discussed in Section 3. It is seen that for all the three cables, the sag affects the fundamental in-plane mode considerably, while its effect on the other modes is negligible, as compared to a taut cable.

3 | LABORATORY TESTING AND THEORETICAL ANALYSIS OF VSDS

3.1 | Viscous-shear dampers

A typical VSD comprises two main parts, that is, a casing box containing viscous medium and a movable part with shearing plates that are submerged in the viscous medium, as illustrated in Figure 2. When the shearing plates move relatively to the casing, a resistance force is generated owing to the high viscous drag of the viscous medium against

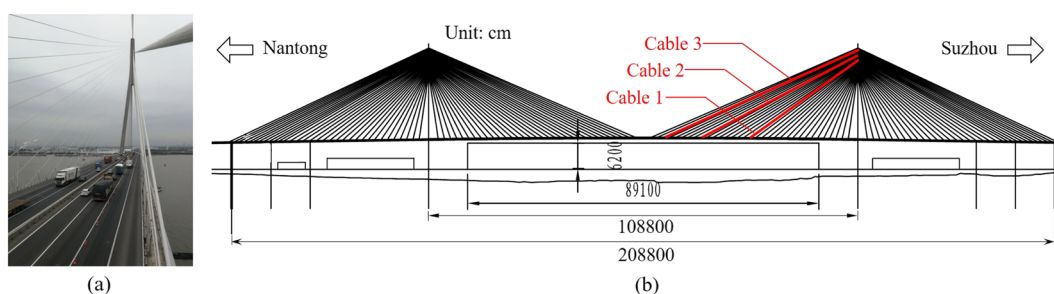
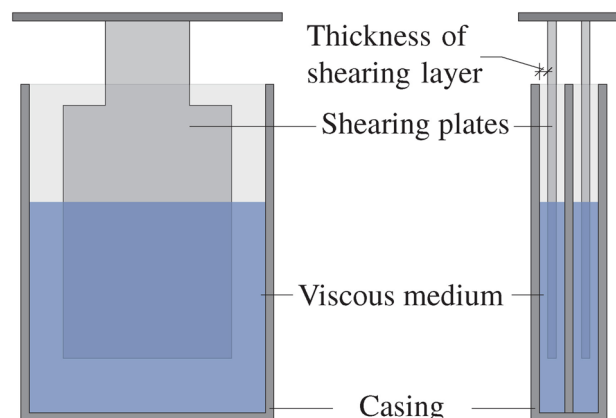


FIGURE 1 The Sutong Bridge: (a) a photo of the south bridge pylon and (b) the elevation plot and the three cables for this study

TABLE 1 Cable parameters and dynamic properties

No.	Length l (m)	Mass m (kg/m)	Diameter D (m)	Tension H (kN)	Cross-section Area (mm ²)	Inclination Angle θ (°)	Sag-extensibility Parameter λ^2	In-plane mode frequency (Hz)		
								Mode 1	Mode 2	Mode 3
1	334.2	64.5	0.128	4,182.5	7,658	34	0.78	0.393	0.762	1.143
2	454.1	77.7	0.139	5,099.0	9,275	27	1.50	0.299	0.564	0.847
3	546.9	91.3	0.151	6,240.5	10,891	23	1.97	0.258	0.478	0.718

**FIGURE 2** A schematic diagram of the viscous-shear damper

shearing. A constant thickness of the viscous layer between the moving plate and side plates of the casing is supposed to be maintained during deformation. The resistance force depends mainly on the viscosity of the medium, the thickness of the viscous layer between shearing plates, and the total shear area.²⁰ As seen in Figure 3, the VSD is simple in structure. Unlike the widely used oil dampers or hydraulic dampers, the VSD does not have any expensive and vulnerable parts like sealing ring, valves, pistons, ball joints, and so on, which makes it low-cost and requires less maintenance effort.

In the design of the VSDs for testing, the thickness of the viscous layer between plates is chosen mainly to prevent possible contact of the two adjacent plates in cable vibrations. In other words, the thickness cannot be too small, to avoid the collision between the neighboring plates when the angle between the deck and the cable changes in response to traffic loading and temperature variation. The angle change is considerable for such a long cable-stayed bridge. The viscous medium was made and then tested for viscosity. The initial shear area was determined using the empirical formulas⁴⁴ providing the viscosity, the thickness of the viscous layer, and the target viscous coefficient which was theoretically optimized for multimode control based on the analytical formulas provided by Krenk and Nielsen¹¹ for a shallow cable with a viscous damper (see also Figure 18). The stiffness of the VSD was initially ignored. For the three cables in Table 1, three VSDs were fabricated and then tested to determine their actual viscous damping coefficient. The shear area of each VSD was then adjusted by comparing the measured coefficients with the target ones. Table 2 gives the

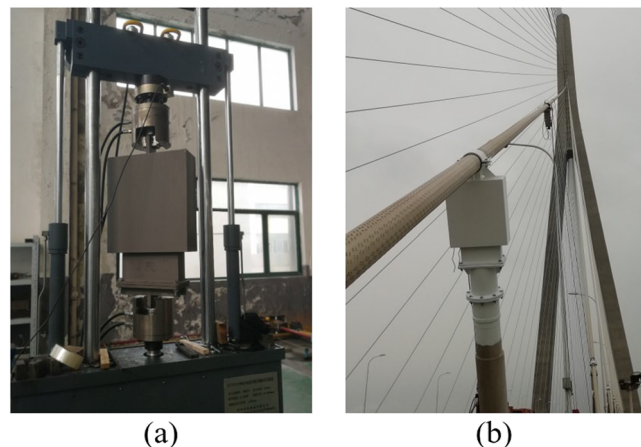
**FIGURE 3** The viscous-shear damper: (a) in laboratory testing and (b) installed between the cable and the bridge deck on the Sutong Bridge. Note that the damper was painted after the installation on the bridge for testing

TABLE 2 Viscous-shear damper parameters

Cable no.	Relative damper position (%)	Thickness of viscous layer (cm)	Viscosity ^a (Pa·s)	Shear area (cm ²)
1	2.60	1.2	15,000	454
2	2.36	1.2	15,000	530
3	2.21	1.2	15,000	630

^aThe viscosity listed was measured at 20°.

parameters of the final VSDs after adjustments. Note that the viscosity of the viscous medium depends on the temperature (see the studies by Yoneda^{20,44}) while the effect of this dependence on the attained cable damping was found to be limited in previous studies.^{20,44} In this study, the VSDs were tested both in laboratory and on the bridge in ambient air with temperature around 20°. A detailed investigation of the temperature effect on the multimode cable vibration control using the VSD is left for future study.

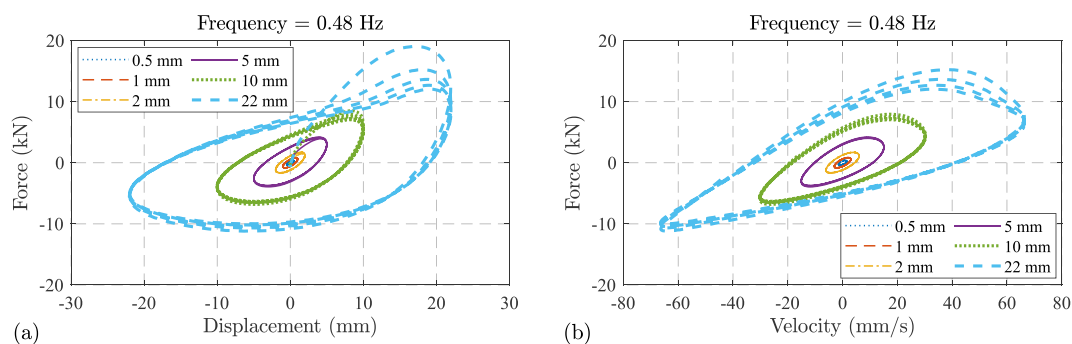
3.2 | Laboratory testing

The designed VSDs were tested on a servo-hydraulic testing machine in a laboratory, as shown in Figure 3a. Forced sinusoidal displacements were applied to each VSD by a 10-kN capacity computer-controlled actuator dynamically. The actual damper deformation and the corresponding resistance force were recorded along with time samples. The sampling frequency of data collection was 100 Hz, and in each test, 10 cycles were performed. Frequencies of the forced displacement were chosen according to the modal frequencies of the corresponding cable, and the test cases are listed in Table 3. The corresponding cable mode index is indicated in the bracket following each frequency. Note that for the first cable mode, the frequency of the forced displacement in the damper testing was close to but not exactly the cable frequency as the sag effect was neglected in the laboratory test design, which has limited effects on the theoretical damping effect prediction. For each frequency, the damper behaviors under five different displacement amplitudes were tested, as also listed in Table 3. A total of 84 tests were performed on the three dampers.

The testing results of the final VSDs are presented as follows. Figure 4a shows curves of the resistance forces and the displacements for VSD 3 under forced displacements with frequency of 0.48 Hz for the five amplitudes. Figure 4b shows the corresponding force-velocity curves. It is seen from Figure 4a that when the shearing plates move towards the casing box, corresponding to positive displacement, the amplitude of the resistance force is larger as compared to

TABLE 3 Viscous-shear damper laboratory testing cases

Cable/damper no.	Frequency of forced displacement (Hz)	Amplitude of forced displacement (mm)
1	0.38 (first), 0.76 (second), 1.52 (fourth), 2.29 (sixth), 3.05 (eighth)	0.5, 1, 2, 5, 10, 16
2	0.28 (first), 0.56 (second), 1.13 (fourth), 3.10 (11th)	0.5, 1, 2, 5, 10, 20
3	0.24 (first), 0.48 (second), 1.20 (fifth), 1.92 (eighth), 3.12 (13th)	0.5, 1, 2, 5, 10, 22

**FIGURE 4** Examples of laboratory testing results of viscous-shear damper 3: (a) force-displacement curves and (b) force-velocity curves

that corresponds to negative displacements. This is because the shear area is increased as more part of the plates are submerged into the viscous medium when the damper is under compression. This leads to asymmetric behaviors of the VSD when experiences compression and extension, particularly when the displacement amplitude is large, as seen in Figure 4. The asymmetric shape of the force-displacement curves is similar to that presented by Yoneda.⁴⁴ In the figure, it is also shown that the resistance force decays in the first several cycles and tends to stabilize after the decay. In the experiments, 10 cycles of forced displacements were found to be sufficient for stabilization.

3.3 | Damper modeling and parameter identification

Figure 4a indicates that the damper has intrinsic stiffness. Therefore, the Kelvin–Voigt model, illustrated in Figure 5, where k_d denotes the damper stiffness and c_d denotes the damping coefficient, is used to approximate dynamic property of the VSD. The resistance force of the VSD is thus expressed as

$$f(t) = k_d v_a(t) + c_d \dot{v}_a(t), \quad (2)$$

where $v_a(t)$ denotes the damper displacement and $\dot{v}_a(t)$ is the corresponding velocity.

For each tested case under a forced displacement with a specific frequency and amplitude, the equivalent damper stiffness and damping coefficient, that is, k_d and c_d , were identified from the recorded displacement and resistance force in the last cycle, using the least squares method. Figure 6 compares the force-displacement and force-velocity curves approximated using the identified Kelvin–Voigt model with the laboratory measurements shown in Figure 4. Note that only the curves corresponding to the last cycle in the experiments are plotted in Figure 6 for comparison. The results in Figure 6 suggest that the Kelvin–Voigt model can better describe the VSD behavior when the damper deformation amplitude is relatively low, that is, no more than 10 mm. For a larger damper deformation amplitude, the modeling error increases because of the aforementioned asymmetric behavior of the VSD.

The identified parameters of the three VSDs are plotted in Figures 7–9 with respect to the displacement amplitude for different frequencies. It is seen that the stiffness and damping coefficients are dependent of the frequency and amplitude of the forced displacement. On variations of the coefficients, the following features are observed.

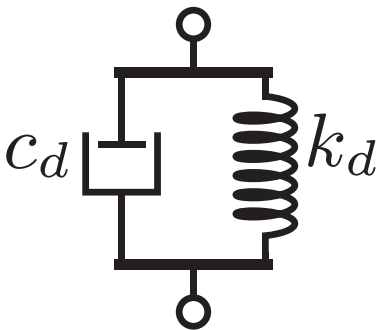


FIGURE 5 The Kelvin–Voigt model for the viscous-shear damper

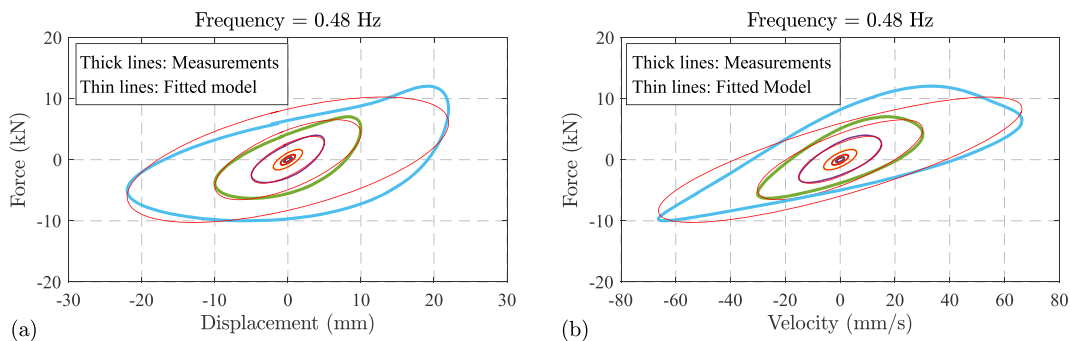


FIGURE 6 Comparison of the testing result and the fitted model for viscous-shear damper 3: (a) force-displacement curves and (b) force-velocity curves

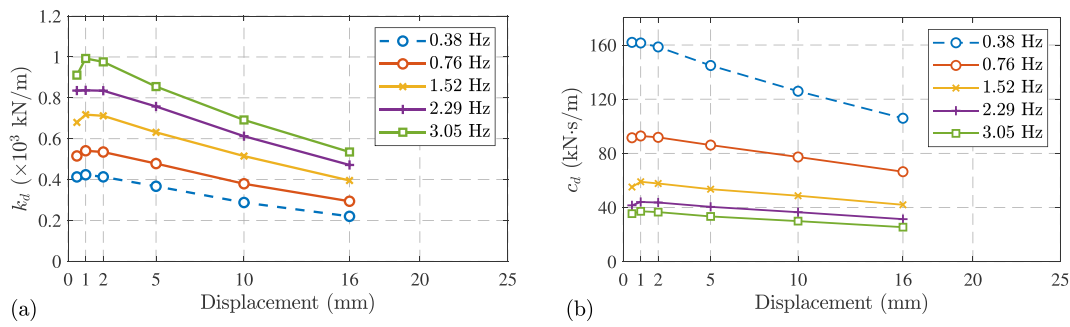


FIGURE 7 Identified mechanical properties of viscous-shear damper 1: (a) stiffness coefficient and (b) damping coefficient

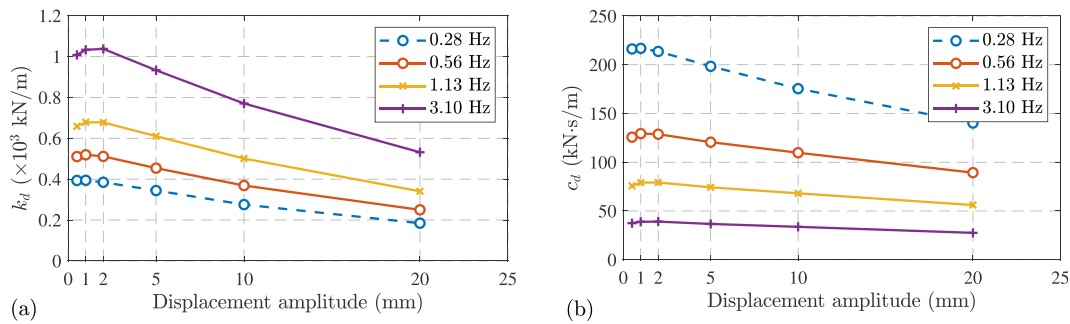


FIGURE 8 Identified mechanical properties of viscous-shear damper 2: (a) stiffness coefficient and (b) damping coefficient

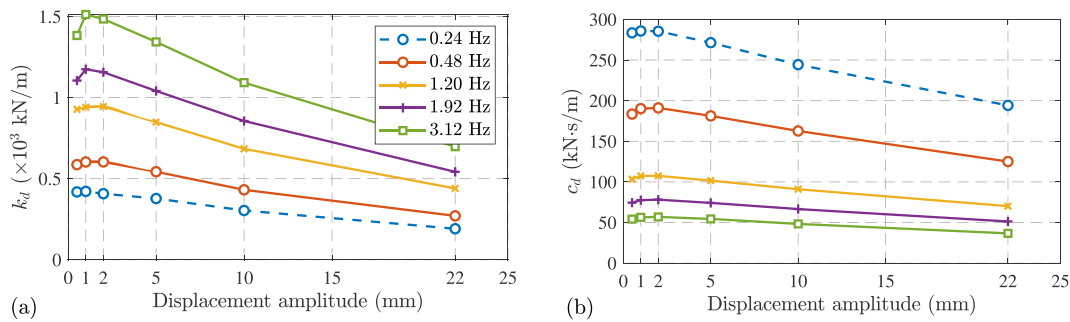


FIGURE 9 Identified mechanical properties of viscous-shear damper 3: (a) stiffness coefficient and (b) damping coefficient

- For a specific amplitude, as the frequency of the forced displacement increases, the damping coefficient decreases while the stiffness coefficient increases.
- For a specific frequency of the forced displacement and when the amplitude is relatively large (larger than 2 mm), both the damping coefficient and the stiffness coefficient decreases as the amplitude increases; while for very small amplitudes, the coefficients decrease slightly as the amplitude decreases.
- For a specific frequency of the force displacement, the damper stiffness decreases faster than the damping coefficient with respect to the increasing amplitude.

Generally, properties of the three VSDs are consistent. With increasing shear area, from VSD 1 to VSD 3, the stiffness and damping coefficients are increased. Note that the damping coefficient is decreasing with respect to the frequency which is desirable for multimode cable vibration control as the optimal damper coefficient decreases for higher mode vibration.^{6,7} However, the increasing stiffness coefficient for higher frequencies reduces the damping effect. The combination of those two effects needs to be addressed. The influences of the frequency-dependent damper properties on the damping effect of taut cables have been considered in previous studies.^{9,19} The displacement dependence of the

damper stiffness and damping coefficients indicates the nonlinearity of the damper behaviors. For a detailed treatment of the damper nonlinearity and its influence on the damping effect to the cable, one may refer to the recent studies.^{37,38} For the convenience of practical design, the linear Kelvin–Voigt model is used for the VSDs while their damping effect on cable is discussed respectively for different displacement amplitudes and frequencies.

3.4 | Analytical damping effect

In response to the presence of intrinsic stiffness of the VSDs and the sag effect of the cables (as indicated in Table 1), the cable-damper system is modeled as a shallow cable with a Kelvin–Voigt damper near one cable support to study the damping effect analytically. Note that Javanbakht et al²³ additionally considered the cable bending stiffness and negative damper stiffness. For long cables, the bending stiffness can be ignored for simplicity. For the completeness of this study, the analytical derivation is briefly introduced in the following.

3.4.1 | Complex modal analysis

To formulate the system characteristic equation, the generalized scheme presented by Krenk and Høgsberg⁹ is followed while the sag effect is additionally included. As illustrated in Figure 10, a cable with sag is considered to be damped by a general transverse force $f(t)$ at a distance of a from the closer cable support. Small sag is assumed, and the shallow cable model is thus appropriate. In other words, the static profile of the cable is assumed to be a parabolic function,

$$y(x) = 4d \left(1 - \frac{x}{l}\right) \frac{x}{l}, \quad (3)$$

where x is the coordinate axis along the cable chord as shown in Figure 10 and $y(x)$ = cable static profile. The sag at the mid span is denoted by d . Dynamic displacement of the cable, denoted by $v(x, t)$ in Figure 10, is governed by the partial differential equation that

$$H \frac{\partial^2 v}{\partial x^2} + h \frac{d^2 y}{dx^2} - m \frac{\partial^2 v}{\partial t^2} = f(t) \delta(x - a), \quad (4)$$

where h denotes the increased cable tension due to cable vibration and $\delta()$ denotes the Dirac delta function.

Denote the cable vibration amplitude at $x = a$ as $\tilde{v}_a$. The general solution of the equation of motion is of the form¹¹:

$$\tilde{v}(x) = \tilde{v}_a \frac{\sin(\beta x)}{\sin(\beta a)} + \frac{8d}{(\beta l)^2 H} \frac{\tilde{h} [1 - \cos(\beta x)] \sin(\beta a) - [1 - \cos(\beta a)] \sin(\beta x)}{\sin(\beta a)}, 0 \leq x \leq a, \quad (5)$$

$$\tilde{v}(x') = \tilde{v}_a \frac{\sin(\beta x')}{\sin(\beta a')} + \frac{8d}{(\beta l)^2 H} \frac{\tilde{h} [1 - \cos(\beta x')] \sin(\beta a') - [1 - \cos(\beta a')] \sin(\beta x')}{\sin(\beta a')}, 0 \leq x' \leq a', \quad (6)$$

where x' denotes the coordinate axis along the cable chord originated from the anchorage farther from the transverse force and correspondingly the distance $a' = l - a$. In the expressions, variables with a tilde denote the amplitude of the

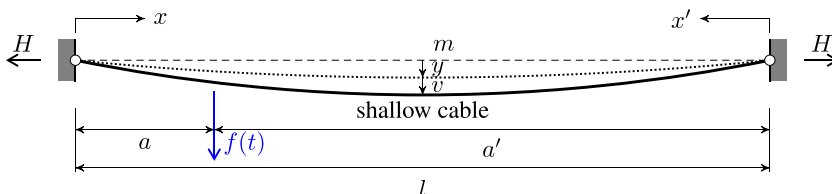


FIGURE 10 A shallow cable under a transverse force

corresponding time-dependent variables, for example, $h(t) = \tilde{h} \exp(i\omega t)$ with $i = \sqrt{-1}$ and ω = damped cable circular frequency. The wavenumber β is related to the frequency ω by

$$\beta = \omega \sqrt{\frac{m}{H}}. \quad (7)$$

The relation accounting for elastic elongation of the cable can be written in the frequency domain, with the time variable $\exp(i\omega t)$ eliminated from both sides, as¹¹

$$\frac{\tilde{h}L_e}{EA} = \frac{mg}{H} \int_0^l \tilde{v}(x) dx = \frac{mg}{H} \left[\int_0^a \tilde{v}(x) dx + \int_0^{a'} \tilde{v}(x') dx' \right], \quad (8)$$

in which $L_e \approx l[1+8(d/l)^2]$ ⁵³ for parabolic cable static profile. As shown by Krenk and Nielsen,¹¹ cable support flexibility can also be included in the definition of λ^2 . Furthermore, the following relation is found that

$$\frac{8\tilde{h}d}{H} = \frac{\lambda^2}{l} \left[\int_0^a \tilde{v}(x) dx + \int_0^{a'} \tilde{v}(x') dx' \right]. \quad (9)$$

Substituting Equations (5) and (6) into the preceding equation, one finds

$$\frac{1}{l} \int_0^a \tilde{v}(x) dx = \frac{\tilde{v}_a}{\beta l} \tan \frac{\beta a}{2} + \frac{8d}{(\beta l)^3 H} \tilde{h} \left(\beta a - 2 \tan \frac{\beta a}{2} \right). \quad (10)$$

Eventually, the relation between $\tilde{h}$ and $\tilde{v}_a$ is obtained as

$$\frac{8d}{(\beta l)^2 H} \tilde{h} \left[2 \tan \frac{\beta a}{2} + 2 \tan \frac{\beta a'}{2} - \beta l + \frac{1}{\lambda^2} (\beta l)^3 \right] = \tilde{v}_a \left(\tan \frac{\beta a}{2} + \tan \frac{\beta a'}{2} \right). \quad (11)$$

The internal force equilibrium at the point of the transverse load can also be written in the frequency domain as

$$\frac{d\tilde{v}(a+)}{dx} - \frac{d\tilde{v}(a-)}{dx} = \frac{\tilde{f}}{H}, \quad (12)$$

with $\tilde{f}$ denotes the amplitude of the transverse force, and $d\tilde{v}(a-)/(dx)$ and $d\tilde{v}(a+)/(dx)$ denote the left and right derivatives of the cable dynamic displacement at $x = a$ and $x' = a'$, respectively. Note that

$$l \frac{d\tilde{v}(a-)}{dx} = \beta l \cot(\beta a) \tilde{v}_a + \frac{8d}{\beta l H} \tilde{h} \tan \frac{\beta a}{2} = \beta l \tilde{v}_a \left[\cot(\beta a) + \frac{\tan^2(\beta a/2) + \tan(\beta a'/2) \tan(\beta a/2)}{2 \tan(\beta a/2) + 2 \tan(\beta a'/2) - \beta l + (\beta l)^3 / \lambda^2} \right]. \quad (13)$$

The characteristic equation of a shallow cable with a generalized linear transverse force is thus written as

$$\cot(\beta a) + \cot(\beta a') + \frac{[\tan(\beta a/2) + \tan(\beta a'/2)]^2}{2 \tan(\beta a/2) + 2 \tan(\beta a'/2) - \beta l + (\beta l)^3 / \lambda^2} = -\frac{\tilde{f}/H}{\beta \tilde{v}_a}. \quad (14)$$

For a taut cable (with sag effect neglected), the third term in the left-hand side of the preceding equation vanishes, and the equation reduces to that derived by Krenk and Høgsberg.⁹

For a damper represented by the Kelvin–Voigt model, its force-deformation relation (2) is expressed in the frequency domain as

$$\tilde{f} = (k_d + i\omega c_d)\tilde{v}_a. \quad (15)$$

Substituting the preceding expression into Equation (14) and after simplification, the characteristic equation becomes

$$\sin \frac{\beta l}{2} \left\{ \sin \frac{\beta l}{2} - \cos \frac{\beta l}{2} \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right] \right\} = -2\alpha \sin \frac{\beta a}{2} \sin \frac{\beta a'}{2} \left\{ \sin \frac{\beta l}{2} - \cos \frac{\beta a}{2} \cos \frac{\beta a'}{2} \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right] \right\}, \quad (16)$$

where the damper parameters are nondimensionalized as

$$\mu = \frac{k_d a}{H}, \quad \eta = \frac{c_d}{\sqrt{mH}}, \quad \alpha = \frac{\mu}{\beta a} + i\eta. \quad (17)$$

Substitution of α in Equation (16) by $i\eta$ leads to the characteristic equation of a shallow cable with a viscous damper as discussed by Krenk and Nielsen.¹¹ The derivation is briefed here. The readers are advised to refer to Krenk and Nielsen¹¹ for details.

Equation (16) can be simplified respectively for nearly symmetric and nearly antisymmetric modes (see Krenk and Nielsen¹¹). For nearly symmetric modes, the equation can be written as

$$\tan \frac{\beta l}{2} = \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right] + \frac{2\alpha \left[\frac{\sin(\beta a/2) \sin(\beta a'/2)}{\sin(\beta l/2)} \right]^2 \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right]^2}{1 + 2\alpha \frac{\sin(\beta a/2) \sin(\beta a'/2)}{\sin(\beta l/2)} \left\{ 1 - \frac{\sin(\beta a/2) \sin(\beta a'/2)}{\sin(\beta l/2)} \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right] \right\}}, \quad (18)$$

and for nearly antisymmetric modes, the following iterative formula is obtained:

$$\tan \frac{\beta l}{2} = \frac{2\alpha \sin^2 \frac{\beta a}{2} \left\{ \sin \frac{\beta l}{2} - \cos \frac{\beta a}{2} \cos \frac{\beta a'}{2} \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right] \right\}}{\sin \frac{\beta l}{2} - \cos \frac{\beta l}{2} \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right] + \alpha \sin(\beta a) \left\{ \sin \frac{\beta l}{2} - \cos \frac{\beta a}{2} \cos \frac{\beta a'}{2} \left[\frac{\beta l}{2} - \left(\frac{\beta l}{2} \right)^3 \frac{4}{\lambda^2} \right] \right\}}. \quad (19)$$

The fixed-point iteration described by Krenk and Nielsen¹¹ is also applicable to solve the preceding two equations. The modal damping ratio, ζ_n , can then be evaluated from the solved complex wave number by

$$\zeta_n = \Im(\beta_n l) / |\beta_n l|, \quad (20)$$

where n is the mode number. Considering that the damper is quite close to one cable support, the following asymptotic solutions are obtained for nearly symmetric modes as

$$\frac{\zeta_n}{a/l} \approx \frac{\left(\tan \frac{\beta_n^0 l}{2} - \frac{\beta_n^0 l}{2} \right)^2}{\tan^2 \left(\frac{\beta_n^0 l}{2} \right) + \frac{12}{\lambda^2} \left(\frac{\beta_n^0 l}{2} \right)^2} \frac{\eta \beta_n^0 a}{(1 + \mu)^2 + (\eta \beta_n^0 a)^2}, \quad n = 1, 3, \dots, \quad (21)$$

where β_n^0 is the wavenumber of the shallow cable without any attachment and can be solved from Equation (18) by letting $\alpha=0$. For nearly antisymmetric modes, one finds

$$\frac{\zeta_n}{a/l} \approx \frac{\eta n \pi a / l}{(1 + \mu)^2 + (\eta n \pi a / l)^2}, n = 2, 4, \dots, \quad (22)$$

as $\beta_n^0 l = n\pi$ in this particular case.

3.4.2 | Analytical prediction using the identified damper parameters

With the damping and stiffness coefficients identified from the laboratory testing, the damping effect provided by the dampers in various situations can then be predicted using Equations (18) and (19) or Equations (21) and (22). As the approximate equations become less accurate for higher modes which are interested herein, Equations (18) and (19) are used for evaluating the damping effect numerically.

As listed in Table 3, the damper for Cable 1 was tested under frequencies when the cable is vibrating in the first, second, fourth, sixth, and eighth modes. In each of these modes, the variation of the logarithmic decrement damping versus the damping and stiffness coefficient is illustrated in Figure 11. Note that the logarithmic decrement damping, denoted by δ_n for the n th cable mode, is related to the modal damping ratio ζ_n by

$$\delta_n = 2\pi\zeta_n. \quad (23)$$

On each contour in the graphs, the combination of the damper stiffness and damping coefficients achieves the same level of damping, and the label indicates the value of the logarithmic decrement. The measured damper stiffness and damping coefficients for different damper deformation amplitudes are indicated in Figure 11 using markers. The damping effect of the damper on the cable can be estimated accordingly, and it is seen for each of the five modes that the logarithmic decrement damping is around 0.03. It is also suggested that the intrinsic stiffness largely reduces the damping effect of the VSDs.

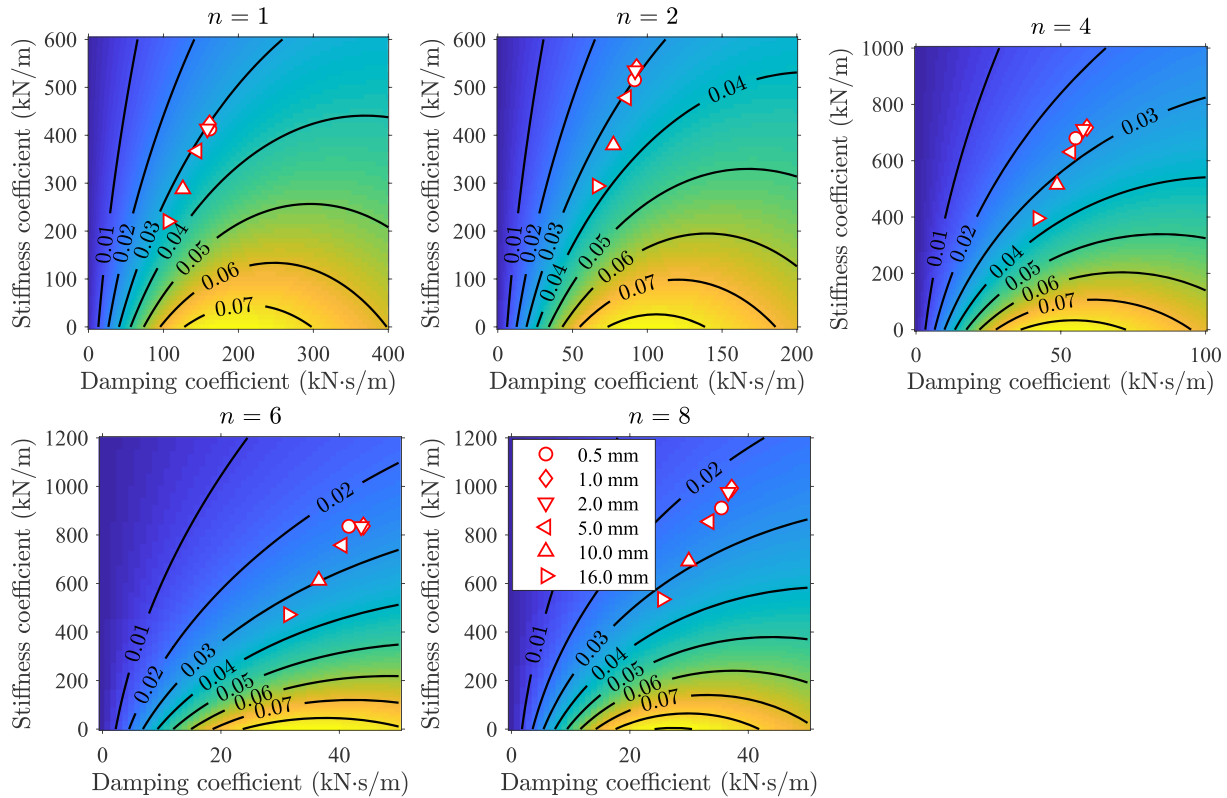


FIGURE 11 Theoretical prediction of the logarithmic decrement damping of Cable 1 with the viscous-shear damper

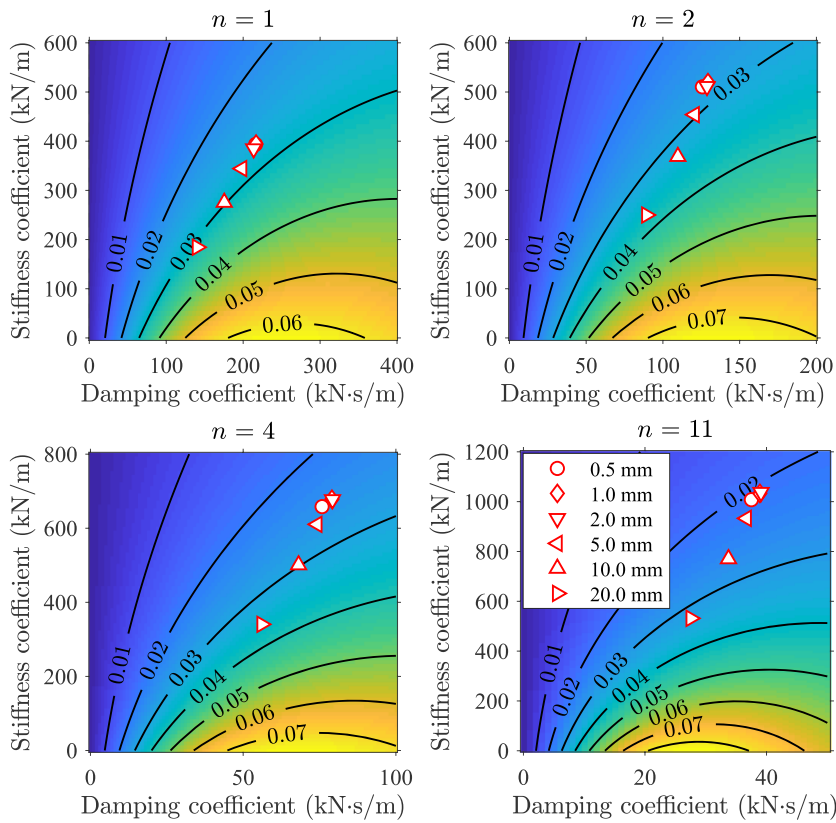


FIGURE 12 Theoretical prediction of the logarithmic decrement damping of Cable 2 with the viscous-shear damper

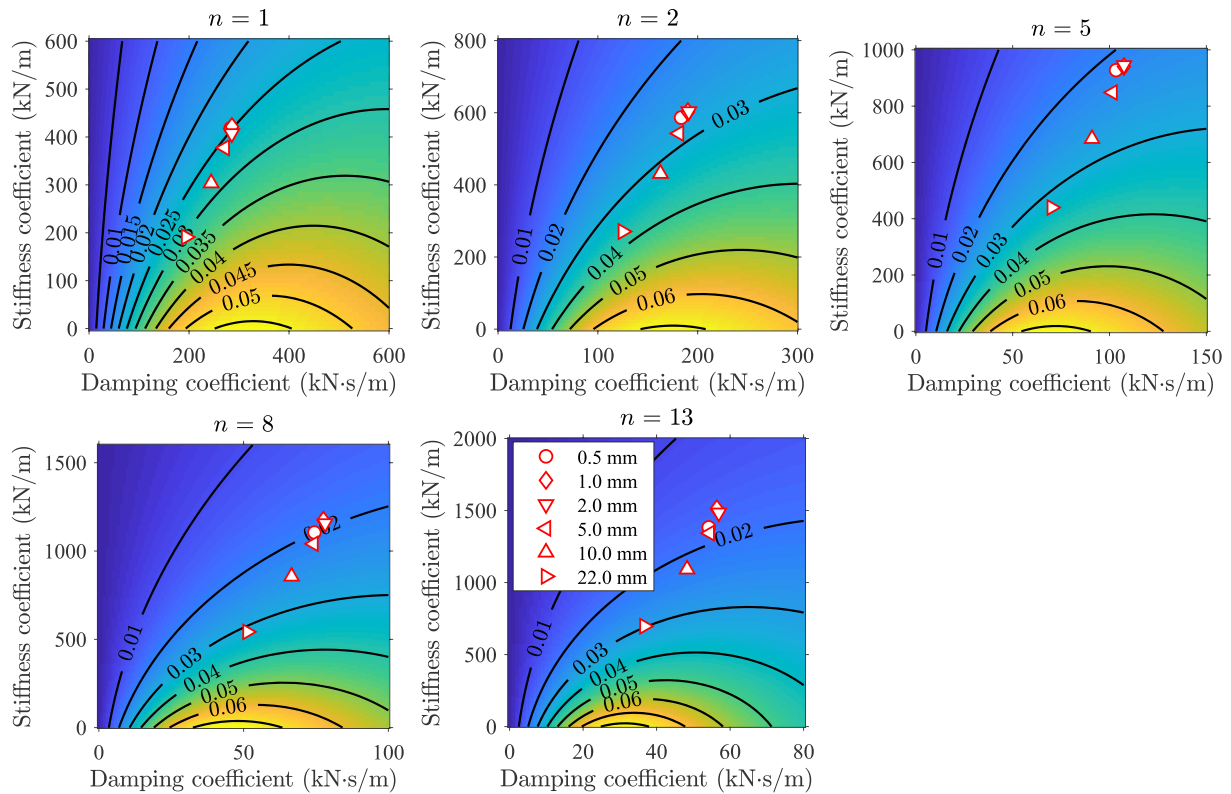


FIGURE 13 Theoretical prediction of the logarithmic decrement damping of Cable 3 with the viscous-shear damper

Figures 12 and 13 show the variation of the logarithm decrement damping with respect to the damper stiffness and damping coefficients for selected modes of Cables 2 and 3, respectively. The measured VSD parameters are indicated accordingly. For the first, second, and fourth modes of Cable 2, the predicted logarithmic decrement is around 0.03, while for the 11th mode, the logarithmic decrement is relatively smaller, between 0.02 and 0.03. For Cable 3, the predicted logarithmic decrement damping of the first, fifth, and eighth modes is between 0.02 and 0.03, while it is around 0.03 for the second mode and it is around 0.02 for the 13th mode.

It is seen from Figures 11 to 13 that, as the damper deformation amplitude increases, the damping effect increases due to an apparent decrement of the damper intrinsic stiffness (recall Figures 7 to 9). On the contrary, the variation of the damping coefficients with respect to the damper deformation amplitude has less influence on the damping effect. This suggests that the VSD performance can be largely improved by reducing its intrinsic stiffness which is notwithstanding quite challenging in practice.

4 | DAMPING EFFECTS EVALUATED BY FIELD TESTING

4.1 | Testing scheme

Field testing was then conducted to measure damping of the cable without any dampers and with the corresponding VSD. The free-decay testing is probably the most reliable method to determine the structural damping, and it was used for this study. In other words, the cable was first excited to vibrate in a target mode, and the vibration was left to decay freely after its amplitude reached a certain value; the modal damping of this mode was then computed from the cable free decay response. The testing scheme is illustrated in Figure 14. Accelerometers were placed on the cable about 13 m above the bridge deck to measure cable accelerations. A displacement sensor was installed to measure the deformation of the VSD. For vibration excitation, an oscillator was attached to the cable also about 13 m above the bridge deck. Steel wires were used to excite the cable vibration manually in lower modes. The sensors and the oscillator were installed with the aid of a cherry picker. Figure 15 illustrates the installation of the sensors and the oscillator.

For each cable or cable-VSD system, effort was made to excite as many modes as possible to study multimode damping effects. As previously mentioned, cable modes with frequency larger than 3 Hz are of less interest for rain-

FIGURE 14 Field testing setup. VSD, viscous-shear damper

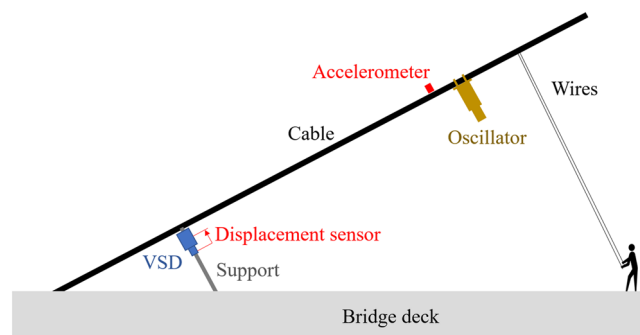
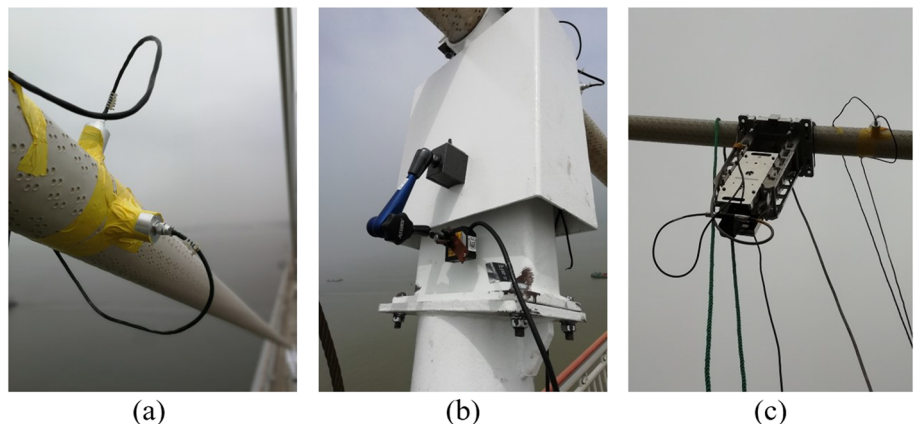


FIGURE 15 Photos of the installed sensors and oscillator in the field testing: (a) accelerometers, (b) laser displacement sensor, and (c) oscillator



wind-induced vibrations and hence only several such modes were tested for demonstration. It was found that for cable modes with frequency larger than 1 Hz, single-mode vibration can be excited successfully by only using the oscillator, and for modes with frequency less but close to 1 Hz, single-mode vibrations can be excited manually with the aid of the oscillator. However, for the first two modes of the tested cables, the oscillator failed to work completely due to the limited installation height and vibrating mass, and manual excitation is also extremely difficult due to the large cable length and the presence of sag. Only for Cable 3, the excitation of the second mode vibration was considered to be acceptable, and the result will be presented for reference. Recall that rain–wind-induced cable responses have frequency mainly in 1 and 3 Hz and thus the first two modes of the long cables are somehow less interesting since the corresponding frequencies are less than 1 Hz as seen in Table 1. It needs to stress that the oscillator was the key enabling the evaluation of multimode damping effects of the VSD which have not been reported in the literature.

4.2 | Testing results

A typical acceleration record of the successfully excited single-mode cable vibration in the field testing is shown in Figure 16a, which corresponds to the sixth mode vibration of Cable 1 with the VSD installed. The Fourier spectrum of the acceleration in the free decay period (when the oscillator was turned off) is shown in Figure 16b, which has a dominant peak corresponding to the frequency of this cable mode. In this case, the excitation was completed solely by the oscillator, and it is pretty successful. Notwithstanding, owing to the ambient excitation from the bridge deck and also from the wind loading as the experiments were conducted when the bridge was in operation, other vibration components also exist as indicated by other peaks in Figure 16b. Therefore, a bandwidth filter with the peak frequency as the center frequency and a width of 0.4 Hz is used to preprocess the acceleration signal, and the filtered signal is shown in Figure 16c. The envelopes of the filtered acceleration in the free decay period are then obtained and used to compute the damping ratio by curve fitting, as illustrated in Figure 16d.

For each case, the same procedure was applied to determine the vibration frequency, the damping ratio, and the corresponding logarithmic decrement damping. The results of the three cables are listed in Tables 4–6, where the results of some modes are missing because the corresponding cases were not tested due to the time limitation. In Table 4, it is seen that for Cable 1, the logarithmic decrement corresponding to the inherent damping is less than 0.01, while with the VSD, the logarithmic decrement damping for modes lower than the eight mode (frequency lower than 3.101 Hz) is larger than 0.03. For higher modes of Cable 1, the damping is slightly decreased. For Cable 2, when without

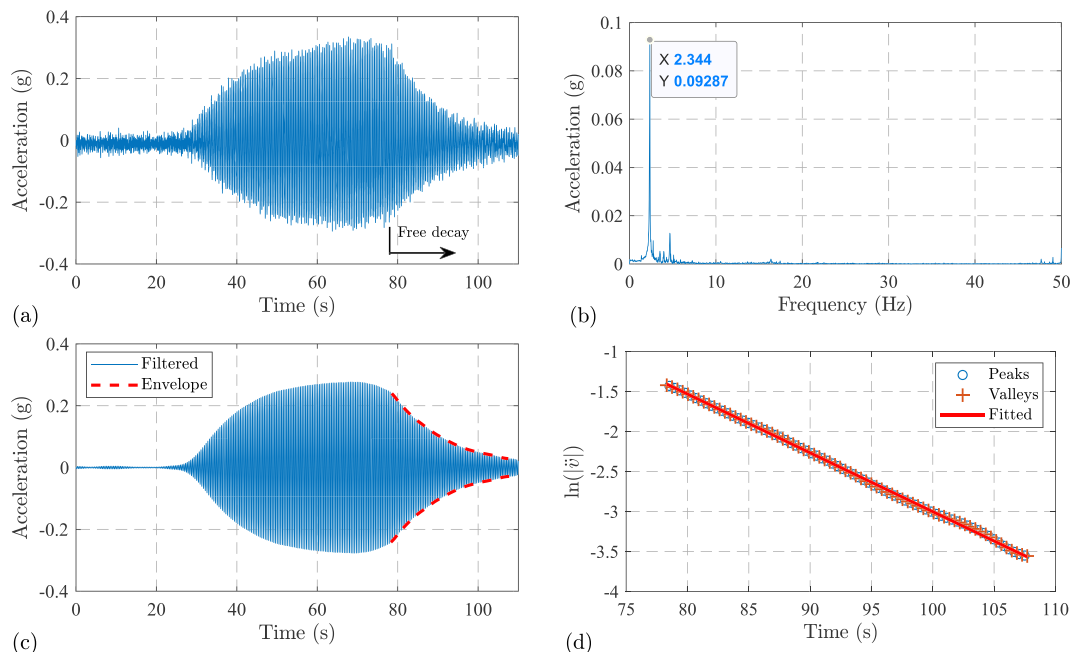


FIGURE 16 Acceleration measurement and data processing procedure for damping evaluation: (a) raw data, (b) Fourier spectrum of the free decay period, (c) filtered acceleration, and (d) curve fitting of the free decay envelopes

TABLE 4 Field testing results of Cable 1 without and with the corresponding VSD

Mode no.	Without VSD			With VSD		
	Freq. (Hz)	Damping ratio (%)	Log. decrement	Freq. (Hz)	Damping ratio (%)	Log. decrement
2	0.763	0.15	0.0092	0.781	0.62	0.0391
3	1.147	0.09	0.0056	1.160	0.53	0.0333
4	1.563	0.07	0.0041	1.563	0.50	0.0314
5	1.950	0.07	0.0046	1.953	0.49	0.0308
6	2.289	0.06	0.0036	2.344	0.50	0.0314
7	2.698	0.05	0.0032	2.710	0.48	0.0302
8	3.094	0.05	0.0030	3.101	0.46	0.0289
9	3.485	0.05	0.0034	—	—	—
10	3.900	0.05	0.0030	3.882	0.44	0.0276
15	—	—	—	5.835	0.39	0.0245

Abbreviation: VSD, viscous-shear damper.

TABLE 5 Field testing results of Cable 2 without and with the corresponding VSD

Mode no.	Without VSD			With VSD		
	Freq. (Hz)	Damping ratio (%)	Log. decrement	Freq. (Hz)	Damping ratio (%)	Log. decrement
2	—	—	—	0.537	1.50	0.0945
3	0.815	0.09	0.0056	0.830	0.52	0.0327
4	1.082	0.04	0.0025	1.123	0.50	0.0314
5	1.350	0.05	0.0031	1.392	0.46	0.0289
6	1.654	0.06	0.0037	1.660	0.46	0.0289
7	1.913	0.04	0.0025	1.929	0.43	0.0270
8	2.185	0.03	0.0019	2.197	0.40	0.0251
9	—	—	—	2.466	0.42	0.0264
10	2.731	0.04	0.0025	2.734	0.36	0.0226
11	—	—	—	3.027	0.40	0.0251
12	3.278	0.07	0.0044	3.320	0.47	0.0295
15	—	—	—	4.199	0.42	0.0264
20	5.457	0.07	0.0044	5.566	0.46	0.0289

Abbreviation: VSD, viscous-shear damper.

dampers, the cable modal damping ratio is less than 0.1% for all the tested modes, and with the VSD, the total modal damping ratio is between 0.36% and 0.52%; the logarithmic decrement damping is larger than 0.03 for the third and fourth modes and is 0.0289 for the fifth mode; for higher modes, the logarithmic decrement is between 0.025 and 0.03. For Cable 3, as in Table 6, the logarithmic decrement when without any dampers is less than 0.01 and decreases with respect to the modal frequency. The logarithmic decrement damping of Cable 3 with the VSD is larger than 0.03 for the second to the fifth mode, and the logarithmic decrement for the second mode is abnormally larger (0.0596) than that of other modes, which is probably attributed to a larger estimation error as the excitation of this mode vibration was not very successful. For higher modes of Cable 3, the total damping is decreasing, while the logarithmic decrement damping is generally larger than 0.02 except for the 13th mode.

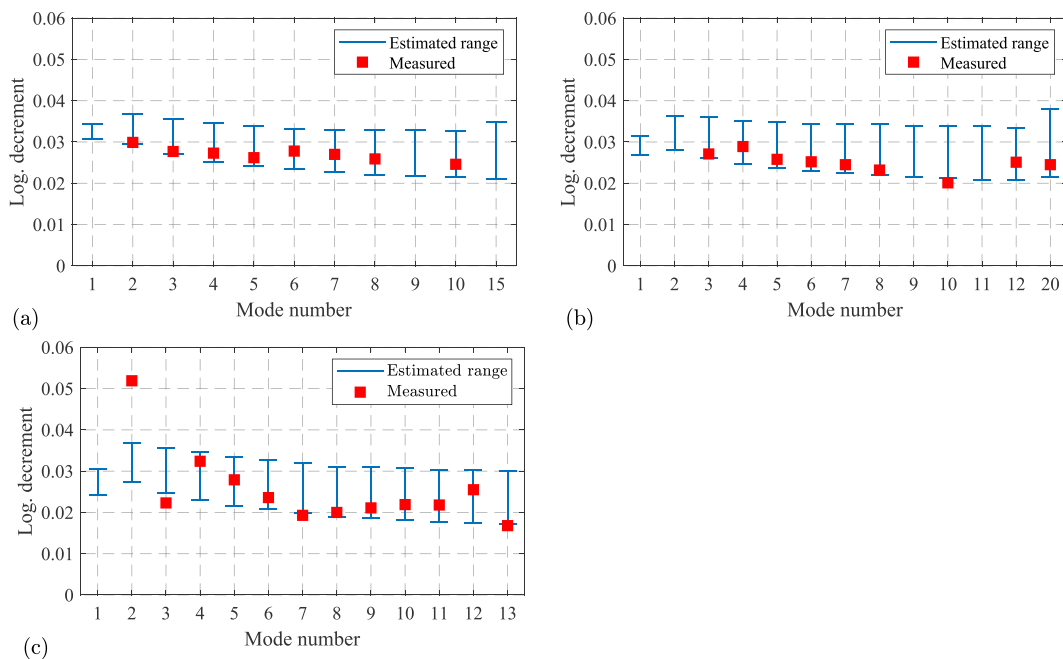
According to the design guideline proposed by the U.S. Federal Highway Administration,¹ the minimum desired Scruton numbers (S_c) for mitigating rain-wind cable vibrations are $m\zeta/(\rho D^2) > 10$ for regular cable arrangements and $m\zeta/(\rho D^2) > 5$ for cables with pipe surface treatments. Note that ρ is the density of air. Using $\rho = 1.225 \text{ kg/m}^3$ and the

TABLE 6 Field testing results of Cable 3 without and with the corresponding VSD

Mode no.	Without VSD			With VSD		
	Freq. (Hz)	Damping ratio (%)	Log. decrement	Freq. (Hz)	Damping ratio (%)	Log. decrement
2	0.473	0.15	0.0097	0.476	0.95	0.0596
3	0.708	0.15	0.0097	0.708	0.51	0.0320
4	0.940	0.13	0.0084	0.952	0.65	0.0408
5	1.178	0.11	0.0066	1.208	0.61	0.0395
6	1.407	0.08	0.0053	1.428	0.46	0.0289
7	1.642	0.08	0.0052	1.685	0.39	0.0245
8	1.880	0.07	0.0045	1.929	0.39	0.0245
9	2.112	0.05	0.0034	2.148	0.39	0.0245
10	2.353	0.05	0.0032	2.393	0.40	0.0251
11	2.588	0.05	0.0033	2.637	0.40	0.0251
12	2.844	0.05	0.0034	2.881	0.46	0.0289
13	3.082	0.03	0.0020	3.101	0.31	0.0195
14	–	–	–	3.345	0.36	0.0224

Abbreviation: VSD, viscous-shear damper.

cable parameters in Table 1, for $m\zeta/(\rho D^2) > 10$, the minimum damping ratios for the three cables are 0.31%, 0.30%, and 0.31%. Generally, the measured total damping ratios of the tested modes of all the cable-VSD systems are larger than or equal to the minimum requirement. To summarize, the inherent modal damping of the tested cables is less than 0.15% and decreases with respect to cable modal frequency (strictly decreasing for Cables 1 and 3); the VSDs performs satisfactorily and the performances of the three dampers are consistent. The damping effects provided by the VSD to lower cable modes are slightly larger than those to higher modes, and generally, they are comparable, indicating the capacity of the VSD for multimode cable vibration control.

**FIGURE 17** Comparison of the measured and the theoretical damping effect: (a) Cable 1, (b) Cable 2, and (c) Cable 3

4.3 | Comparison with analytical predictions

To compare the field testing results with analytical estimates using the method in Section 2, the VSD provided damping is computed from Tables 4 to 6 by subtracting cable inherent damping of the cable from the total damping, assuming that they are linearly superposed. Note that the dampers were tested only under four or five frequencies as discussed in Section 3, while in field testing, more modes were tested. Therefore, the stiffness and damping coefficients for each damper under a specific deformation amplitude were approximated as an exponential function of the modal frequency (exponential function was also used by Yoneda²⁰) respectively, which was then used to predict the damper parameters when the cable vibrates in other modes. With the interpolated/extrapolated damper parameters for a particular mode, Equations (18) or (19) can then be used to compute the corresponding damping effects for varied damper deformation amplitudes, leading to a range of the modal damping for each mode of the cable.

The predicted ranges in terms of the logarithmic decrement are plotted in Figure 17 versus the cable mode number, for Cables 1 to 3 with the VSD, respectively. The measured logarithmic decrement damping is indicated by a red square in the figures for each mode. Clearly, for nearly all the modes, the measurement is in the range of the analytical prediction. One large deviation is found in the second mode of Cable 3. As mentioned before, this is probably attributed to the fact that the second mode vibration was not excited very successfully in the field test and hence the damping evaluation is not as accurate as that of other modes. Notice that the lower bound of the predicted damping effect for each mode corresponds to the case when the cable vibrates and hence the damper deforms with small amplitude (damper deformation amplitude less than 2 mm) because of the dominant effect of the large damper stiffness. This finding is valuable for practical design since the VSD deformation amplitude in operation is unknown and changing while the

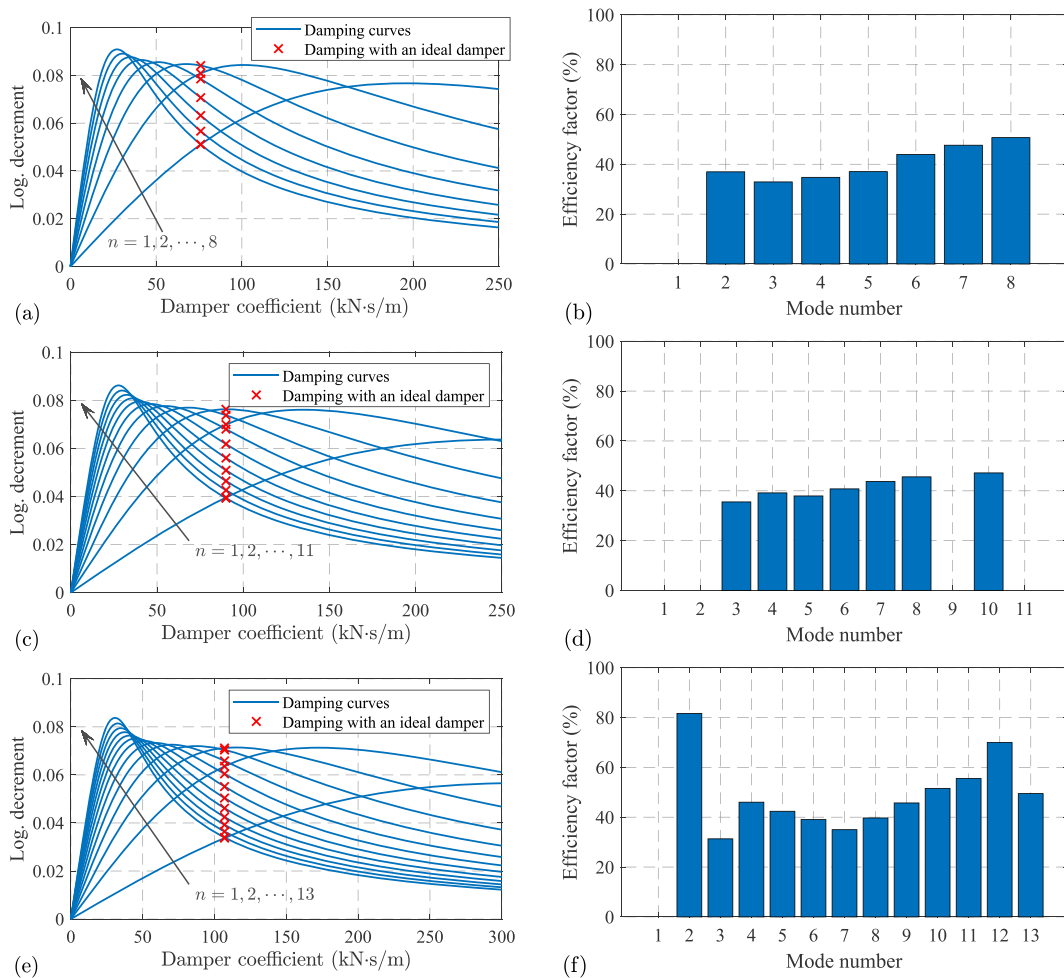


FIGURE 18 Efficiency factors of the VSDs: (a) damping effects of Cable 1 with an ideal viscous damper, (b) efficiency factors of VSD 1 on Cable 1, (c) damping effects of Cable 2 with an ideal viscous damper, (d) efficiency factors of VSD 2 on Cable 2, (e) damping effects of Cable 3 with an ideal viscous damper, and (f) efficiency factors of VSD 3 on Cable 3. VSD, viscous-shear damper

analytical analysis using the VSD properties tested under small-amplitude deformation provides the lower bound of the overall damping effect.

In previous studies, for example, Sun et al.,¹⁵ the efficiency factor of a damper is defined as the ratio between the actual damping and the maximal attainable damping by an ideal viscous damper, that is, $0.5a/l$ in terms of modal damping ratio. However, considering to use an ideal damper with a constant coefficient for multimode vibration control, the maximal attainable damping for each mode is different. Here, the coefficient of the ideal damper is optimized to maximize the lower bound of the damping ratios for all the modes with frequency less or close to 3 Hz. Therefore, the first 8, 11, and 13 modes are considered for Cables 1, 2, and 3, respectively, as seen from Tables 4 to 6. Figure 18a, c, and e shows damping curves of the three cables with an ideal damper at the corresponding damper position given in Table 2. The optimal damper coefficients according to the aforementioned multimode control criterion are 76, 90, and 107 kN·s/m, respectively. The damping effects on all the concerned modes of each cable attached with the optimal damper are indicated using crosses in the figures. The efficiency factors are then computed as the ratio between the measured supplemental damping by the VSD and the corresponding damping attained by the optimized ideal damper for each mode. The efficiency factors are illustrated in Figure 18b, d, and f, respectively for VSD 1 to VSD 3. It is seen that for most modes, the efficiency factors are between 30% and 50%, while for the second and 12th modes of Cable 3, the efficiency factors are abnormal probably due to relatively large measurement errors. The efficiency factors tend to increase with respect to the mode number. The efficiency is generally acceptable for practice, and it can be largely improved if the damper stiffness is reduced. Notably, the efficiency factors of the VSDs are comparable to those of the oil dampers on a full-scale cable,¹⁵ while the VSD is low-cost and low-maintenance as compared to the oil damper and hence is advantageous for multimode cable vibration control. Note the fact that the damping coefficient of the VSD decreases with respect to frequency improves its performance in cable multimode vibration control.

5 | CONCLUSIONS

This paper presents a comprehensive experimental study of multimode cable vibration control using a VSD, along with theoretical analysis. Three VSDs have been fabricated and optimized respectively for three long cables of the Sutong Bridge. They were tested in laboratory, and their dynamic properties were modeled as a viscous dashpot in parallel with a linear spring while both the stiffness and damping coefficients depend on the damper deformation frequency and amplitude. The VSDs were then installed respectively on the corresponding cable, and free-decay tests were conducted to appreciate the cable damping mode by mode. The measurement results have been further compared with the analytical results from complex modal analysis of the system of a shallow cable attached with a Kelvin–Voigt damper. The following conclusions can be drawn from the analytical and experimental results.

1. The VSD, although has a simple structure, can provide satisfactory multimode control effects for long cables in cable-stayed bridges, and the efficiency factors are mostly between 30% and 50% as compared to an ideal viscous damper optimized for multimode cable vibration control.
2. The VSD can be modeled as a Kelvin–Voigt damper for analytical analysis of the cable-damper system while the amplitude- and frequency-dependence of the coefficients need to be considered. The analytical predictions are found to be consistent with the field measurements, and hence, the laboratory testing together with the analytical analysis method presented here can be used for VSD design in practice. Particularly, the lower bound of the damping effect provided by a VSD for each cable mode can be determined using the tested damper properties under small-amplitude deformation.
3. The damping coefficient of the VSD decreases with respect to damper deformation frequency, which is favorable in multimode cable vibration control.
4. It is the relatively large damper stiffness responsible for the reduction of the damping effect, and future VSD design may be directed to reducing its inherent stiffness.

Note that here, the effect of temperature on the VSD's performance in multimode cable vibration control has not been analyzed. This will be studied in future by laboratory testing of the VSDs in lower temperatures in combination with the theoretical prediction method presented in this study.

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